

Ranjeet Kumbhar

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Research Interests

My research interests broadly include Applied Artificial Intelligence, Machine Learning, Deep Learning, Exoskeletons and Robotics. I am keen to explore new research directions and integrate cutting-edge techniques across disciplines to advance intelligent systems – and ultimately, build AI that works in the real world.

Education

Master of Engineering in Computer Science & Engineering

Thapar Institute of Engineering & Technology, Patiala

Aug 2024 – Jul 2026

CGPA: 9.0 / 10

Thesis: *AI-Enabled Adaptive Controller for Lower-Limb Exoskeleton Synchronous Motion in Healthy Elderly Individuals*

- Implemented model-free adaptive control and reinforcement learning for sit-to-stand and walking gait tasks.
- Developed and validated high-fidelity robot dynamic simulations using OpenSim, MATLAB/Simulink.

Bachelor of Engineering in Information Technology (Honours: AI/ML)

Savitribai Phule Pune University, Pune

Jan 2021 – May 2024

CGPA: 9.36 / 10

Publications

R. Kumbhar, R. Singh, A. M. Gadade, D. R. Kumar, I. Hussain. “Sensitivity-Driven Deep Learning Model for Tribological Prediction in Al7075/B₄C Nanocomposites”, **Journal of Materials Research and Technology**, (Elsevier, SCIE, Q1, IF: 7.3), 2026. DOI: 10.1016/j.jmrt.2026.04.196

R. Kumbhar, A. M. Gadade, R. Singh, M. Choudhary. “Tribo-Informatics Method for Predictive Modeling of Friction and Wear Mechanisms in Al7075/B₄C Metal Matrix Composites”, **Journal of Materials Engineering and Performance**, (Springer, SCIE, Q2, IF: 2.6), 2026. DOI: 10.1007/s11665-026-13723-3

R. Kumbhar, R. Singh, A. M. Gadade, A. Singla, I. Hussain. “A Novel Hybrid Controller for Sit-to-Stand Assistance using a Simscape Model based Lower-Limb Exoskeleton”, **Proc. Int. Conf. Modelling, Identification and Control (ICMIC)**, 2026. DOI: 10.48550/arXiv.2604.03766

R. Kumbhar, A. M. Gadade, R. Singh, A. Singla, R. Kumar. “Deep RL-Tuned Model-Free Adaptive Control for Lower-Limb Exoskeletons During Sit-to-Stand Transitions”, **ASME Journal of Mechanisms and Robotics (Under Review)**, 2026. DOI: 10.48550/arXiv.2606.22040

Research Experience

Graduate Student Researcher, Thapar Institute of Engineering & Technology, Patiala, India Aug 2025 – Jul 2026

- Conducted research on AI-enabled adaptive control for lower-limb exoskeletons (Master’s Thesis).
- Authored multiple first-author SCIE journal papers and international conference papers in AI, and robotics.

Research Intern, College of Military Engineering (Indian Army), Pune, India

Dec 2022 – Mar 2023

- Designed a blockchain-based secure file-sharing system deployed in a private distributed LAN environment.

- Implemented cryptographic protocols ensuring end-to-end data integrity and permissioned multi-user access control.

Projects

CataractAI – Cataract Detection System Using Deep Learning

[GitHub]

- Medical AI system using computer vision on fundus images for cataract detection, achieving over 90% accuracy.
- Implemented EfficientNetB0 with transfer learning (ImageNet pre-training).

Technical Skills

Programming	Python, C, C++, SQL, HTML/CSS
ML / DL	Deep Learning, Computer Vision, NLP, Statistical Learning, Sensitivity Analysis
Core CS	Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS
Frameworks	MATLAB/Simulink, TensorFlow, PyTorch, Scikit-learn, OpenSim

Achievements & Scholarships

GATE 2024	Data Science & Artificial Intelligence, Score: 422 All India Rank: 4003
AICTE & TIET	Postgraduate Scholarship — INR 3,82,800
SBIF Asha Scholarship	Postgraduate Scholarship — INR 3,20,000
Stanford Online / Coursera	Machine Learning Specialization; TensorFlow Developer Specialization

References

Dr. Ashish Singla Professor, Dept. of Mech. Eng. Thapar Inst. Eng. & Tech., Patiala ashish.singla@thapar.edu	Dr. Rajmeet Singh Mechatronics Engineer, Research Office MBZUAI, UAE rajmeet.bhourji@mbzuai.ac.ae	Dr. Ravinder Kumar Associate Professor, Dept. of Comp. Sci. & Eng. Thapar Inst. Eng. & Tech., Patiala ravinder@thapar.edu
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